The Method of Moments

Theorem 1

Let $m_X(t)$ and $m_Y(t)$ denote the moment-generating functions of random variables X and Y , respectively.

If both moment-generating functions exist and

$$m_X(t) = m_Y(t)$$

for all values of t, then X and Y have the same probability distribution.

Theorem 2

Let Y_1, Y_2 ,..., Y_n be independent random variables with moment generating functions $m_{Y_1}(t), mY_2(t), \ldots, m_{Y_n}(t)$, respectively. If U = $Y_1 + Y_2 + \cdots + Y_n$, then $m_U(t) = m_{Y_1}(t) * m_{Y_2}(t) * \ldots * m_{Y_n}(t)$.

PROOF

We know that, because the random variables $Y_1, Y_2, \ldots, Y_n$ are independent,

$$m_U(t) = E\left[e^{t(Y_1+\cdots+Y_n)}\right] = E\left(e^{tY_1}e^{tY_2}\cdots e^{tY_n}\right)$$
$$= E\left(e^{tY_1}\right) \times E\left(e^{tY_2}\right) \times \cdots \times E\left(e^{tY_n}\right).$$

Thus, by the definition of moment-generating functions,

$m_U(t) = m_{Y_1}(t) * mY_2(t) * \ldots * mY_n(t)$.

Theorem 3

Let $Y_1, Y_2, \ldots, Y_n$ be independent normally distributed random variables with $E(Y_i) = \mu_i$ and $V(Y_i) = \sigma_i^2$, for $i = 1, 2, \ldots, n$, and let $a_1, a_2, \ldots, a_n$ be constants. If

$$U = \sum_{i=1}^{n} a_i Y_i = a_1 Y_1 + a_2 Y_2 + \cdots + a_n Y_n,$$

then U is a normally distributed random variable with

$$E(U) = \sum_{i=1}^{n} a_i \mu_i = a_1 \mu_1 + a_2 \mu_2 + \cdots + a_n \mu_n$$

and

$$V(U) = \sum_{i=1}^{n} a_i^2 \sigma_i^2 = a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + \cdots + a_n^2 \sigma_n^2.$$

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$$V(U) = \sum_{i=1}^{n} a_i^2 \sigma_i^2 = a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + \cdots + a_n^2 \sigma_n^2.$$

Method

Let U be a function of the random variables Y_1, Y_2 ,..., Y_n.

1. Find the moment-generating function for U, m_U (t).
2. Compare m_U (t) with other well-known moment-generating functions. If m_U (t) = m_V (t) for all values of t, Theorem1 implies that U and V have identical distributions.

EXAMPLE 09

Suppose that Y is a normally distributed random variable with mean μ and variance σ^2 .

Show that

$$Z = \frac{(Y - \mu)}{\sigma}$$

has a standard normal distribution, a normal distribution with mean 0 and variance 1.

EXAMPLE 10

Let Z be a normally distributed random variable with mean 0 and variance 1. Use the method of moment-generating functions to find the probability distribution of Z^2 .

Multivariable Transformations Using Jacobians

Method

Suppose that Y_1 and Y_2 are continuous random variables with joint density function $f_{Y1,Y2}(y_1, y_2)$ and that for all (y_1, y_2), such that $f_{Y1,Y2}(y_1, y_2) > 0$,

$$u_1 = h_1(y_1, y_2) \text{ and } u_2 = h_2(y_1, y_2)$$

is a one-to-one transformation from (y_1, y_2) to (u_1, u_2) with inverse

$$y_1 = h_1^{-1}(u_1, u_2) \text{ and } y_2 = h_2^{-1}(u_1, u_2).$$

If $h_1^{-1}(u_1, u_2)$ and $h_2^{-1}(u_1, u_2)$ have continuous partial derivatives with respect to u_1 and u_2 and Jacobian

Cont.

$$J = \det \begin{bmatrix} \frac{\partial h_1^{-1}}{\partial u_1} & \frac{\partial h_1^{-1}}{\partial u_2} \\ \frac{\partial h_2^{-1}}{\partial u_1} & \frac{\partial h_2^{-1}}{\partial u_2} \end{bmatrix} = \frac{\partial h_1^{-1}}{\partial u_1}\frac{\partial h_2^{-1}}{\partial u_2} - \frac{\partial h_2^{-1}}{\partial u_1}\frac{\partial h_1^{-1}}{\partial u_2} \neq 0,$$

then the joint density of U_1 and U_2 is

$$f_{U_1,U_2}(u_1, u_2) = f_{Y_1,Y_2}\left(h_1^{-1}(u_1, u_2), h_2^{-1}(u_1, u_2)\right)|J|,$$

, where |J | is the absolute value of the Jacobian.

EXAMPLE 12

Let Y_1 and Y_2 be independent exponential random variables, both with mean $\beta > 0$. Find the density function of

$$U = \frac{Y_1}{Y_1 + Y_2}.$$

If $Y_1,\ Y_2, \ldots, Y_k$ are jointly continuous random variables and

$U_1 = h_1(Y_1,\ Y_2, \ldots, Y_k),\ U_2 = h_2(Y_1,\ Y_2, \ldots, Y_k), \ldots,\ U_k = h_k(Y_1,\ Y_2, \ldots, Y_k),$

where the transformation

$$u_1 = h_1(y_1, y_2, \ldots, y_k), u_2 = h_2(y_1, y_2, \ldots, y_k), \ldots, u_k = h_k(y_1, y_2, \ldots, y_k)$$

is a one-to-one transformation from $(y_1, y_2, \ldots, y_k)$ to $(u_1, u_2, \ldots, u_k)$ with inverse

$$y_1 = h_1^{-1}(u_1, u_2, \ldots, u_k),\ y_2 = h_2^{-1}(u_1, u_2, \ldots, u_k), \ldots,$$
$$y_k = h_k^{-1}(u_1, u_2, \ldots, u_k),$$

and $h_1^{-1}(u_1, u_2, \ldots, u_k)$, $h_2^{-1}(u_1, u_2, \ldots, u_k), \ldots, h_k^{-1}(u_1, u_2, \ldots, u_k)$ have continuous partial derivatives with respect to $u_1,\ u_2, \ldots,\ u_k$ and *Jacobian*

$$J = \det \begin{bmatrix} \dfrac{\partial h_1^{-1}}{\partial u_1} & \dfrac{\partial h_1^{-1}}{\partial u_2} & \cdots & \dfrac{\partial h_1^{-1}}{\partial u_k} \\[2ex] \dfrac{\partial h_2^{-1}}{\partial u_1} & \dfrac{\partial h_2^{-1}}{\partial u_2} & \cdots & \dfrac{\partial h_2^{-1}}{\partial u_k} \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\partial h_k^{-1}}{\partial u_1} & \dfrac{\partial h_k^{-1}}{\partial u_2} & \cdots & \dfrac{\partial h_k^{-1}}{\partial u_k} \end{bmatrix} \neq 0,$$

EXAMPLE 13

Let Z_1 and Z_2 be independent standard normal random variables and $U_1 = Z_1$ and $U_2 = Z_1 + Z_2$.

a Derive the joint density of U_1 and U_2 .
b $E(U_1)$, $E(U_2)$, $V(U_1)$, $V(U_2)$, and $Cov(U_1, U_2)$.
c Are U_1 and U_2 independent? Why?

Order Statistics

- Many functions of random variables of interest in practice depend on the relative magnitudes of the observed variables.
- we often order observed random variables according to their magnitudes. The resulting ordered variables are called order statistics.

let Y_1, Y_2 ,..., Y_n denote independent continuous random variables with distribution function $F(y)$ and density function $f(y)$.
We denote the ordered random variables Y_i by $Y(1), Y(2),\ldots, Y(n)$, where $Y(1) \le Y(2) \le \cdots \le Y(n)$.

- $Y(1) = \min(Y_1, Y_2, \ldots, Y_n)$ is the minimum of the random variables Y_i, and $Y(n) = \max(Y_1, Y_2, \ldots, Y_n)$ is the maximum of the random variables Y_i.

- The probability density functions for $Y(1)$ and $Y(n)$ can be found using the method of distribution functions.

Density function of $Y_{(n)}$

- $Y_{(n)}$ is the maximum of $Y_1, Y_2, \ldots, Y_n$, the event $(Y(n) \le y)$ will occur if and only if the events $(Y_i \le y)$ occur for every $i = 1, 2, \ldots, n$.
- That is, $P(Y_{(n)} \le y) = P(Y_1 \le y, Y_2 \le y, \ldots, Y_n \le y)$.
- Because Y_i are independent and $P(Y_i \le y) = F(y)$ for $i = 1, 2, \ldots, n$, it follows that the distribution function of $Y_{(n)}$ is given by

$$F_{Y_{(n)}}(y) = P(Y_{(n)} \le y) = P(Y_1 \le y)P(Y_2 \le y) \cdots P(Y_n \le y) = [F(y)]^n.$$

- Letting $g_{(n)}(y)$ denote the density function of $Y(n)$, we see that, on taking derivatives of both sides,

$$g_{(n)}(y) = n[F(y)]^{n-1} f(y).$$

Density function for $Y_{(1)}$

- The distribution function of $Y_{(1)}$ is

$$F_{Y_{(1)}}(y) = P(Y_{(1)} \leq y) = 1 - P(Y_{(1)} > y).$$

- Because Y(1) is the minimum of Y1, Y2 ,..., Yn, it follows that the event (Y(1) > y) occurs if and only if the events (Yi > y) occur for i = 1, 2 ,..., n.
- Because the Yi are independent and P(Yi > y) = 1 − F(y) for i = 1, 2 ,..., n, we see that

$$\begin{aligned} F_{Y_{(1)}}(y) &= P(Y_{(1)} \leq y) = 1 - P(Y_{(1)} > y) \\ &= 1 - P(Y_1 > y, Y_2 > y, \ldots, Y_n > y) \\ &= 1 - [P(Y_1 > y)P(Y_2 > y)\cdots P(Y_n > y)] \\ &= 1 - [1 - F(y)]^n. \end{aligned}$$

- Thus, if g(1)(y) denotes the density function of Y(1), differentiation of both sides of the last expression yields

$$g_{(1)}(y) = n[1 - F(y)]^{n-1} f(y).$$

EXAMPLE 14

Electronic components of a certain type have a length of life Y , with probability density given by

$$f(y) = \begin{cases} (1/100)e^{-y/100}, & y > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

(Length of life is measured in hours.) Suppose that two such components operate independently and in series in a certain system (hence, the system fails when either component fails). Find the density function for X, the length of life of the system.

EXAMPLE 15

Suppose that the components in Example 13 operate in parallel (hence, the system does not fail until both components fail). Find the density function for X, the length of life of the system.